Review: Energy & Momentum

Answer key

Conventions used throughout

• **Simple harmonic motion (Lesson 03):** restoring force $\vec{F} = -k\vec{x}$, period of a mass on a spring $T = 2\pi\sqrt{(m/k)}$, amplitude A = the largest displacement from equilibrium.

• **Energy:** $E_e = \frac{1}{2}kx^2$, $E_k = \frac{1}{2}mv^2$.

• **Momentum and impulse (Lesson 07):** $\vec{p} = m\vec{v}$ and $\vec{F}\Delta t = \Delta\vec{p} = m\vec{v}_2 - m\vec{v}_1$.

• **Proportionality statements:** "$\propto$" means "is proportional to". Everything else in the equation is held constant.

• **Solve in letters first:** each relationship is written as a formula before it is described.

CONCEPT QUESTION 1

Relationships between variables

What this key covers: the textbook questions listed at the top of the worksheet (pages 274–281, 248 and 266) are not printed on the worksheet, so this key answers the three Concept Questions only.

1. State the relationship (if any) between the following sets of variables. If possible, write a proportionality statement.

   a) period and frequency

   b) acceleration and displacement in SHM

   c) period and force constant for a mass on a spring in SHM

   d) the maximum speed of a body in SHM and the amplitude of its motion.

For each pair, write the equation that links the two variables, then read the proportionality from it.

a) period and frequency

The period is the time for one cycle and the frequency is the number of cycles per second.

Formula: $T = \frac{1}{f}$

Proportionality: $T \propto \frac{1}{f}$

They are inversely proportional: doubling the frequency halves the period.

b) acceleration and displacement in SHM

The only force is the restoring force. Use Newton's second law.

Formula: $F_{net} = ma \quad \Rightarrow \quad -kx = ma$

Divide both sides by m: $a = -\frac{k}{m}x$

Proportionality: $a \propto -x$

The size of the acceleration is directly proportional to the displacement, and it always points the opposite way (back toward equilibrium).

c) period and force constant for a mass on a spring in SHM

Formula: $T = 2\pi\sqrt{\frac{m}{k}}$

Proportionality: $T \propto \frac{1}{\sqrt{k}}$

The period is inversely proportional to the square root of the force constant: a spring four times as stiff gives half the period.

d) the maximum speed of a body in SHM and the amplitude of its motion.

All of the elastic energy at the amplitude becomes kinetic energy at the equilibrium position.

Formula: $\frac{1}{2}kA^2 = \frac{1}{2}mv_{max}^2$

Multiply by $\frac{2}{m}$, square root: $v_{max} = A\sqrt{\frac{k}{m}}$

Proportionality: $v_{max} \propto A$

The maximum speed is directly proportional to the amplitude: doubling the amplitude doubles the maximum speed.

FINAL ANSWER

a) T = 1/f, so T ∝ 1/f (inverse)

b) a = −(k/m)x, so a ∝ −x (direct, opposite direction)

c) T = 2π√(m/k), so T ∝ 1/√k

d) v_{max} = A√(k/m), so v_{max} ∝ A (direct)

CONCEPT QUESTION 2

Conditions for SHM

2. State the condition(s) necessary for SHM to occur.

There must be a **restoring force**: a net force that always points back toward the equilibrium position and whose size is directly proportional to the displacement from equilibrium, $\vec{F} = -k\vec{x}$. For the motion to keep the same amplitude there must also be no friction or other force removing energy (otherwise it is damped harmonic motion).

FINAL ANSWER

A restoring force proportional to the displacement and opposite in direction ($\vec{F} = -k\vec{x}$), with no friction

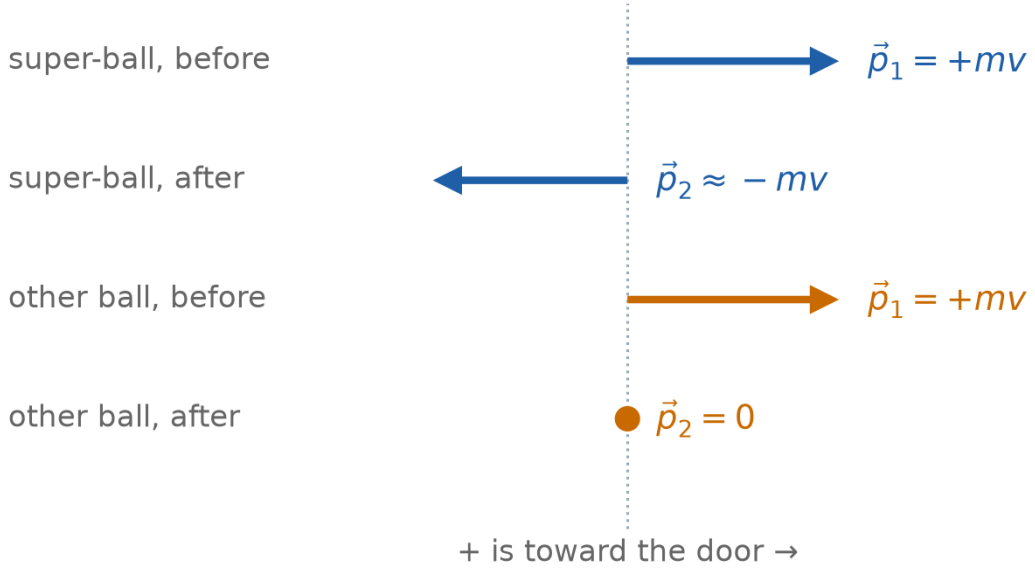
CONCEPT QUESTION 3

Which ball closes the door?

3. You are in your house minding your own business when you hear a funny noise coming from your bedroom. You sneak upstairs and look from the hallway only to see that the monsters you have been afraid of since childhood have come out from under your bed and are running rampant. Not wanting to get too close to the room, you decide to get the door closed by throwing a ball at it. You see two balls of equal mass nearby – one is a super-ball that is very bouncy and the other is made of a material that won't bounce off the door at all, it will just fall to the floor. Which one should you throw? Explain.

Throw both balls with the same velocity +v toward the door and compare the change in momentum of each ball during its collision with the door.

Momentum of each ball before and after it hits the door



The super-ball bounces back at almost the same speed. The other ball stops.

Formula: $\Delta p = mv_2 - mv_1$

Super-ball: $\Delta p \approx m(-v) - m(+v) = -2mv$

Other ball: $\Delta p = m(0) - m(+v) = -mv$

The door gives the super-ball about twice the impulse. By Newton’s third law the ball gives the door an equal and opposite impulse, so the super-ball pushes the door about twice as hard for the same throw.

FINAL ANSWER

Throw the super-ball: bouncing back doubles its change in momentum, so it gives the door about twice the impulse